CHAPTER 7

Antiderivatives and differential equations

S52, S53, S55, S58

Objectives

Calculate antiderivatives; solve a differential equation with an initial condition; interpret an Euler approximation.

Course notes

Antiderivatives on an interval

At this stage we accept that every continuous function on an interval has antiderivatives; the link with area will be established in chapter 8. An antiderivative of f on an interval I is a differentiable function F such that $F' = f$ on I. If F is an antiderivative, all the antiderivatives on I are $F + C$, $C \in \mathbb{R}$.

Proof. If $F' = G' = f$, then $(F - G)' = 0$ on the interval; $F - G$ is therefore constant. Conversely, $(F + C)' = f$. The interval hypothesis is essential: on a disconnected domain, the constants may differ from one interval to another.

Table of antiderivatives — to be read as the reverse of derivatives

$f(x)$	$F(x)$
x^n $(n \in \mathbb{Z},\ n \neq -1)$	$\frac{x^{n+1}}{n+1}$
$\frac{1}{x}$	$\ln \lvert x \rvert$ on an interval avoiding 0
$\frac{1}{\sqrt{x}}$	$2\sqrt{x}$ $(x > 0)$
e^x	e^x
$\cos x$	$\sin x$
$\sin x$	$-\cos x$

The additive constant can always be added. The last two rows will be used again after the study of trigonometry.

Recognising a composition

For a differentiable function u, the antiderivatives of $u'u^n$, $u'e^u$, $\frac{u'}{u}$ are respectively $\frac{u^{n+1}}{n+1}$ $(n \neq -1)$, e^u, $\ln \lvert u \rvert$, on the intervals where the expressions make sense. Linearity allows us to handle sums.

Example. An antiderivative of $6x(x^2+1)^2$ is $(x^2+1)^3$; an antiderivative of e^{3x} is $\frac{e^{3x}}{3}$. To impose $F(0) = 2$ in the latter case, take $F(x) = \frac{e^{3x}}{3} + \frac{5}{3}$.

Differential equations

A differential equation relates an unknown function to its derivatives. A solution must be checked on a specified interval.

Standard solutions

$y' = ay$ has solutions $y(x) = Ce^{ax}$. For $a \neq 0$, $y' = ay + b$ has solutions

$$y(x) = Ce^{ax} - \frac{b}{a}.$$

If $a = 0$, the equation becomes $y' = b$ and $y(x) = bx + C$. If g is a known particular solution of $y' = ay + f(x)$, the solutions are $y = g + Ce^{ax}$.

Proof for $y' = ay$. If y is a solution, $z(x) = e^{-ax}y(x)$ satisfies $z' = e^{-ax}(y' - ay) = 0$; $z = C$. Conversely, the derivative of Ce^{ax} is aCe^{ax}. For $y' = ay + b$, subtracting the constant solution $-\frac{b}{a}$ reduces to the homogeneous equation. For a variable right-hand side, do the same with g.

Example. $y' = -2y + 6$ and $y(0) = 1$ give $y = 3 + Ce^{-2x}$, then $C = -2$. The solution $3 - 2e^{-2x}$ increases towards the equilibrium 3 as $x \to +\infty$.

Euler's method

For $y' = ay + b$, with a step $h > 0$, we approximate $y(t_k)$ by Y_k according to $t_{k+1} = t_k + h$, $Y_{k+1} = Y_k + h(aY_k + b)$. This is an approximation using the tangent, not an equality on the exact solution. A smaller step can improve the approximation; check the error numerically on a known solution.

Research activity

Cooling of a drink — 40 min.

A drink initially at 80 °C is placed in a room kept at 20 °C. For $t \geq 0$, let $T(t)$ be its temperature in degrees Celsius, where t is the elapsed time in minutes. We accept the model $T'(t) = -0.1(T(t) - 20)$ and $T(0) = 80$.

1. Solve this initial value problem and give the expression of $T(t)$.
2. Study the direction of variation of T and find its limit at $+\infty$. Interpret the results.
3. Find the first instant at which the temperature is 40 °C. Give the exact value, then an approximate value in minutes.
4. Apply two steps of Euler's method with a step of 1 minute. Compare the approximation obtained after two minutes with the exact value $T(2)$.
5. Give a concrete situation in which the assumption of a constant room temperature would no longer be appropriate.

Progressive exercises

E1 · Practice. Find all the antiderivatives on $\mathbb{R}$ of the function $x \mapsto 3x^2 - 4x + 1$.

E2 · Practice. Find the antiderivative F of the function $x \mapsto \frac{2x}{x^2+4}$ on $\mathbb{R}$ that satisfies $F(0) = 0$.

E3 · Practice. Find the solution on $\mathbb{R}$ of the differential equation $y' = 3y$ that satisfies $y(1) = 2$.

E4 · Practice. Find the solution on $\mathbb{R}$ of the differential equation $y' = -y + 4$ that satisfies $y(0) = 7$.

E5 · Practice. Consider the differential equation $(E) : y' = y - x$ on $\mathbb{R}$. Check that $g(x) = x + 1$ is a solution of it, then find the solution of (E) that satisfies $y(0) = 0$.

Textbook or past-paper work

Barbazo, p. 228, exercises 25 and 30. In S55: exercise 30, to bring out a constant solution. At home: exercise 25 in full, distinguishing a condition on f from a condition on f'.

Homework

HW7 — An evolution with variable input — 60 min.

We look for a function y defined and differentiable on $[0;+\infty[$ such that $y'(t) = -y(t) + 2t + 1$ and $y(0) = 3$.

1. Find two real numbers a and b such that $g(t) = at + b$ is a solution of the differential equation.
2. Deduce the expression of the solution that satisfies $y(0) = 3$.
3. Study the variations of this solution and state its minimum and the instant at which it is attained.
4. Write a Python function with argument N, a strictly positive integer, that applies Euler's method on $[0;2]$ with step $\frac{2}{N}$ and returns the approximation of $y(2)$. Compare with the exact value for $N = 10$, then $N = 100$.
5. Why is checking the differential equality at only a few instants not enough to prove that a function is a solution?